

SUYOUN KIM

Senior Applied Scientist ◊ Amazon AGI ◊ Full-duplex Speech-to-Speech LLMs

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SUMMARY

Research scientist in speech and multimodal LLMs, working on full-duplex speech-to-speech models that listen and speak at the same time. Conversation has been the thread throughout: from end-to-end conversational speech recognition (Ph.D., Carnegie Mellon) to streaming ASR and spoken language understanding at Meta, to turn-taking, evaluation, and reinforcement learning for full-duplex models at Amazon AGI. Publications at INTERSPEECH, ICASSP, ACL, NAACL, EMNLP, SLT, and IEEE JSTSP, and workshops at NeurIPS and ICLR.

EXPERIENCE

Amazon, AGI

May 2025 – Present

Senior Applied Scientist

Seattle, WA

- Full-duplex speech-to-speech foundation models: a single model that listens and speaks at the same time, and has to decide when to take the floor, when to yield, and when to stay quiet.
- Reinforcement learning for conversational behavior: designed GRPO reward functions for turn-taking, barge-in, and pause handling, with tiered failure-mode rewards and frame-level credit assignment that resist reward hacking, cutting latency while improving answer quality.
- Led evaluation science: owned the end-to-end evaluation infrastructure, metric design, and the evaluation methodology for full-duplex conversational fluidity (turn-taking, barge-in, pause handling, backchanneling).

Meta, GenAI and Reality Labs

Dec 2019 – Apr 2025

Senior Research Scientist

Seattle, WA

- Streaming ASR: LM fusion and contextual biasing for streaming RNN-T.
- ASR metrics: introduced Semantic Distance, an ASR metric for downstream understanding and user-perceived quality.
- On-device spoken language understanding for Meta’s smart glasses assistant.
- GenAI: multimodal LLMs, including contributions to Llama 4.

Abridge AI, Speech Team

Summer 2018

Research Intern

Pittsburgh, PA

- End-to-end speech recognition on medical conversations.

Microsoft Research, Speech and Dialog Research Group

Summer 2017

Research Intern

Redmond, WA

- Language-universal end-to-end speech recognition, with Mike Seltzer, Jinyu Li, and Rui Zhao.

Mitsubishi Electric Research Laboratories (MERL), Speech & Audio Lab

Summer 2016

Research Intern

Cambridge, MA

- Joint CTC/attention end-to-end speech recognition, with Shinji Watanabe and Takaaki Hori.

Carnegie Mellon University

2012 – 2019

Graduate Research Assistant

Pittsburgh, PA

- End-to-end conversational speech recognition (Ph.D.); computational biology, protein–protein interaction (M.S.).

Samsung Electronics, Visual Display Division
Software Engineer

2005 – 2012
Suwon, Korea

- IP set-top box software (audio, video, bootloader) on embedded Linux.

EDUCATION

Carnegie Mellon University 2019
Ph.D. in Electrical and Computer Engineering. Advisors: Richard M. Stern and Florian Metze
Thesis: *End-to-End Speech Recognition on Conversations*. Committee: Florian Metze, Richard M. Stern, Bhiksha Raj, Michael L. Seltzer, and Shinji Watanabe

Carnegie Mellon University 2014
M.S. in Computer Science, Language Technologies Institute. Advisor: Madhavi Ganapathiraju

Georgia Institute of Technology 2011
M.S. in Computer Science (dual degree with Korea University)

Konkuk University 2005
B.S. in Engineering

PUBLICATIONS

Full list: scholar.google.com/citations?user=8tnCOZoAAAAJ

Chun-Yi Kuan, Siwon Kim, Byeonggeun Kim, **Suyoun Kim**, Bo-Ru Lu, Qingming Tang, Ankur Gandhe, Hung-yi Lee, Chieh-Chi Kao, Chao Wang, “Improving Text-to-Audio Instruction Following via Fine-Grained Feedback from Audio-Aware Large Language Models”, (*in INTERSPEECH, 2026*).

Wonjune Kang, Junteng Jia, Chunyang Wu, Wei Zhou, Egor Lakomkin, Yashesh Gaur, Leda Sari, **Suyoun Kim**, Ke Li, Jay Mahadeokar, Ozlem Kalinli, “Frozen Large Language Models Can Perceive Paralinguistic Aspects of Speech”, (*in arXiv, 2024*).

Trang Le, Daniel Lazar, **Suyoun Kim**, Shan Jiang, Duc Le, Adithya Sagar, Aleksandr Livshits, Ahmed Aly, Akshat Shrivastava, “PRoDeliberation: Parallel Robust Deliberation for End-to-End Spoken Language Understanding”, (*in Findings of EMNLP, 2024*).

Zhe Liu, **Suyoun Kim**, Ozlem Kalinli, “Evaluating Speech Recognition Performance Towards Large Language Model Based Voice Assistants”, (*in INTERSPEECH, 2024*).

Roshan Sharma, **Suyoun Kim**, Daniel Lazar, Trang Le, Akshat Shrivastava, Kwanghoon Ahn, Piyush Kansal, Leda Sari, Ozlem Kalinli, Michael Seltzer, “Augmenting text for spoken language understanding with Large Language Models”, (*in SynData4GenAI Workshop, 2024*).

Suyoun Kim, Akshat Shrivastava, Duc Le, Ju Lin, Ozlem Kalinli, Michael L. Seltzer, “Modality Confidence Aware Training for Robust End-to-End Spoken Language Understanding”, (*in INTERSPEECH, 2023*).

Derek Xu, Shuyan Dong, Chaghan Wang*, **Suyoun Kim***, Zhaojiang Lin*, Akshat Shrivastava, Shang-Wen Li, Liang-Hsuan Tseng, Alexei Baevski, Guan-Ting Lin, Hung-yi Lee, Yizhou Sun, Wei Wang, “Introducing Semantics into Speech Encoders”, (*in ACL, 2023*).

Suyoun Kim, Ke Li, Lucas Kabela, Rongqing Huang, Jiedan Zhu, Ozlem Kalinli, Duc Le, “Joint Audio/Text Training for Transformer Rescorer of Streaming Speech Recognition”, (*in Findings of EMNLP, 2022*).

Suyoun Kim, Duc Le, Weiyi Zheng, Tarun Singh, Abhinav Arora, Xiaoyu Zhai, Christian Fuegen, Ozlem Kalinli, and Michael L Seltzer, “Evaluating User Perception of Speech Recognition System Quality with Semantic Distance Metric”, (*in INTERSPEECH, 2022*). [selected to give the oral presentation]

Duc Le, Akshat Shrivastava, Paden Tomasello, **Suyoun Kim**, Aleksandr Livshits, Ozlem Kalinli, Michael L Seltzer, “Deliberation Model for On-Device Spoken Language Understanding”, (*in INTERSPEECH, 2022*). [selected to give the oral presentation]

Suyoun Kim, Abhinav Arora, Duc Le, Ching-Feng Yeh, Christian Fuegen, Ozlem Kalinli, and Michael L Seltzer, “Semantic Distance: A New Metric for ASR Performance Analysis Towards Spoken Language Understanding”, (*in INTERSPEECH, 2021*).

Duc Le, Mahaveer Jain, Gil Keren, **Suyoun Kim**, Yangyang Shi, Jay Mahadeokar, Julian Chan, Yuan Shangguan, Christian Fuegen, Ozlem Kalinli, Yatharth Saraf, and Michael L Seltzer, “Contextualized Streaming End-to-End Speech Recognition with Trie-Based Deep Biasing and Shallow Fusion”, (*in INTERSPEECH, 2021*).

Suyoun Kim, Yuan Shangguan, Jay Mahadeokar, Antoine Bruguier, Christian Fuegen, Michael L Seltzer, and Duc Le, “Improved Neural Language Model Fusion for Streaming Recurrent Neural Network Transducer”, (*in ICASSP, 2021*).

Chunxi Liu, Frank Zhang, Duc Le, **Suyoun Kim**, Yatharth Saraf, and Geoffrey Zweig, “Improving RNN transducer based ASR with auxiliary tasks”, (*in SLT, 2021*).

Suyoun Kim, Siddharth Dalmia, and Florian Metze, “Cross-Attention End-to-End ASR for Two-Party Conversations”, (*in INTERSPEECH, 2019*).

Suyoun Kim, Siddharth Dalmia, and Florian Metze, “Gated Embeddings in End-to-End Speech Recognition for Conversational-Context Fusion”, (*in ACL, 2019*).

Suyoun Kim, and Florian Metze, “Acoustic-to-Word Models with Conversational Context Information”, (*in NAACL, 2019*).

Suyoun Kim, and Florian Metze, “Dialog-context aware end-to-end Speech Recognition”, (*in SLT, 2018*).

Suyoun Kim*, Siddharth Dalmia*, and Florian Metze, “Situation Informed End-to-End ASR for CHiME-5 Challenge”, (*in CHiME Workshop, 2018*).

Suyoun Kim, Michael L. Seltzer, Jinyu Li, and Rui Zhao, “Improved Training for Online End-to-End Speech Recognition Systems”, (*in INTERSPEECH, 2018*).

Suyoun Kim, and Michael L. Seltzer, “Towards Language-universal End-to-End Speech Recognition”, (*in ICASSP, 2018*). [selected to give the oral presentation]

Shinji Watanabe, Takaaki Hori, **Suyoun Kim**, John R Hershey, and Tomoki Hayashi, “Hybrid CTC/Attention Architecture for End-to-End Speech Recognition”, (*IEEE Journal of Selected Topics in Signal Processing, 2017*).

Suyoun Kim, and Ian Lane, “End-to-End Speech Recognition with Auditory Attention for Multi-Microphone Distance Speech Recognition”, (*in INTERSPEECH, 2017*).

Suyoun Kim, Takaaki Hori, and Shinji Watanabe, “Joint CTC-Attention based End-to-End Speech Recognition using Multi-task Learning”, (*in ICASSP, 2017*). [selected to give the oral presentation]

Suyoun Kim, and Ian Lane, “Recurrent Models for Auditory Attention in Multi-Microphone Distant Speech Recognition”, (*in INTERSPEECH, 2016*).

Suyoun Kim, Bhiksha Raj, and Ian Lane, “Environmental Noise Embeddings for Robust Speech Recognition”, (*in arXiv, 2016*).

Suyoun Kim, and Ian Lane, “Recurrent Models for Auditory Attention in Multi-Microphone Distance Speech Recognition,” (*ICLR Workshop, 2016*).

Seungwhan Moon, **Suyoun Kim**, and Haohan Wang, “Multimodal Transfer Deep Learning with an Application in Audio-Visual Recognition” (*NIPS Workshop, 2015*).

AWARDS

Center for Machine Learning and Health Fellowship in Digital Health, Carnegie Mellon (2017 – 2018)

Samsung Graduate Fellowship (2010 – 2011)

TEACHING

Teaching Assistant, Speech Recognition and Understanding (Profs. Ian Lane and Florian Metze), CMU, Fall 2015

Teaching Assistant, Estimation, Detection and Learning (Prof. Rohit Negi), CMU, Spring 2018

Last updated: Oct. 2026